



Asteroid Albedo and Size Inversion: A Deep Learning Method and Application Based on Multiparameters

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Abstract

Of nearly 1.5 million asteroids discovered to date, only about 5% have spectroscopic observations, resulting in limited physical property coverage. Multicolor photometry provides taxonomic classifications for the majority of asteroids. Meanwhile, asteroids originating from the same collisional event and occupying nearby regions of orbital space typically share similar surface properties. Consequently, orbital parameters can provide statistical constraints on asteroid type and albedo, enabling more reliable estimates of their sizes. To achieve accurate inversion of asteroid size and albedo using readily available information under optical observation and to improve the completeness of asteroid physical property databases, this paper proposes a deep learning method that takes an 11-dimensional vector comprising absolute magnitude, orbital parameters, and type information as input, and embeds the Bowell formula to fuse physical priors with data-driven modeling. The proposed method eliminates the dependence on spectral data, thereby significantly expanding the training sample size, which in turn ensures robust model generalization. Test results on asteroids explored by spacecraft or radar observations show mean absolute percentage errors of 26.4% and 21.6% for albedo and effective diameter, respectively. By applying the model to all asteroids with known types, a large-scale asteroid physical properties catalog was constructed. The resulting catalog contains taxonomic type, geometric albedo, and effective diameter for 188,737 asteroids. Compared to existing catalogs based on machine learning predictions, the sample coverage has improved by an order of magnitude. The generated catalog is available on Zenodo ([doi:10.5281/zenodo.20032293](https://doi.org/10.5281/zenodo.20032293)).

Unified Astronomy Thesaurus concepts: Asteroids (72); Astronomy databases (83); Neural networks (1933)

Materials only available in the online version of record: machine-readable table

1. Introduction

Asteroids preserve primitive records of solar system evolution and serve as key tracers for understanding terrestrial planet formation (F. E. DeMeo & B. Carry 2014). Their rich mineral resources also offer potential support for humanity's sustainable development (D. G. Andrews et al. 2015). Taxonomic type, geometric albedo, and size are the key fundamental physical properties. Asteroids located in different regions of the solar system exhibit differences in thermal histories, degrees of space weathering (B. Clark et al. 2023), and composition (R. Binzel et al. 2019), with their types and albedo serving as indicators that reveal these characteristics (L. E. McGraw et al. 2024). Size is a primary parameter for investigating asteroid collisional cascades and assessing the impact hazard posed by near-Earth asteroids (NEAs; W. F. Bottke et al. 2005; A. F. Cheng et al. 2023). Importantly, these three properties are tightly interrelated, as accurate size estimates rely on precise geometric albedo, while albedo can be effectively constrained through taxonomic classification (M. Mahlke et al. 2022; H. Pentikäinen et al. 2026). In addition, the rich orbital parameters, which encode information on formation location and history, can supply additional

statistical information on albedo, thereby supporting reliable size estimation.

A comprehensive catalog of asteroid properties is essential for advancing scientific research and supporting engineering missions, yet the current coverage of asteroid physical parameters remains notably incomplete (J. Berthier et al. 2023). Thermal–physical modeling is one of the primary methods for accurately determining asteroid diameters and albedos from infrared observations (J. R. Masiero et al. 2011; M. Delbo et al. 2015), but such data remain scarce. In the optical band, asteroid diameter estimates typically use the Bowell relation (E. Bowell et al. 1989), which incorporates albedo and absolute magnitude (H) under the assumption of sphericity. However, this method requires adopting an albedo value, such as the average albedo for each taxonomic type, which is discrete and thus constitutes a major source of uncertainty. Moreover, traditional approaches struggle to integrate additional information without substantially increasing analytical complexity, particularly when including statistical constraints from orbital parameters (E. Bowell et al. 1989; A. W. Harris 1998).

Deep learning, with its powerful feature extraction, non-linear mapping, and efficient inference capabilities, excels at integrating high-dimensional vectors in a data-driven manner to infer target parameters. J. Ge et al. (2024, 2025) proposed new deep learning–based methods to combine asteroid reflectance spectra and physical properties (e.g., rotation periods) with orbital parameters (e.g., inclination) for asteroid classification and subsequent albedo and diameter inversion.

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Their experiments demonstrate that this approach can effectively improve parameter inversion accuracy and is readily applicable to processing large-scale data from new sources. By applying the trained model to Gaia Data Release 3 (Gaia DR3; Gaia Collaboration et al. 2023) samples, they constructed a catalog of fundamental physical properties for 58,168 asteroids.

However, spectral data remain indispensable for these previous models. In reality, among the nearly 1.5 million asteroids discovered to date, only about 5% have available spectroscopy (J. Berthier et al. 2023). Most small asteroids and NEAs lack spectral observations due to their low brightness, and nearly 70% of the asteroids with known taxonomic types are classified instead from multiband photometric data (J. Ge et al. 2025). This substantially limits both the applicability of existing models and the scale of the resulting catalogs. In addition, the dependence on parameters such as rotational period, which are missing for many small bodies, also restricts the size of the training dataset and ultimately constrains the generalization performance of deep learning models.

To achieve accurate predictions of albedo and size that approach the results of thermal infrared inversion using optical observations in the absence of spectral data, and to build a more comprehensive catalog of fundamental asteroid properties, this study proposes a deep learning-based method for the joint inversion of continuous asteroid albedo and diameter. By integrating orbital elements with other readily accessible information that provides potential constraints, we trained a highly generalizable model on a substantially larger dataset and then applied it to predict parameters for all asteroids with known types after evaluating its performance. Based on the model outputs and the collected taxonomic results, we construct and release an asteroid physical properties catalog containing 188,737 objects, including their types, albedos, and diameters, for use in related studies or as a comparative reference.

The remainder of this paper is organized as follows. Section 2 describes the dataset and the model for spectroscopy-free inversion of asteroid albedo and diameter. Section 3 presents the validation and test results of the proposed method. Section 4 introduces the constructed asteroid physical properties catalog. Finally, we conclude the paper in Section 5.

2. Data and Method

2.1. Dataset Construction

Photometric and astrometric observations are common methods for studying asteroids and allow the determination of their absolute magnitudes and orbital parameters. A subset of asteroids also has multiband photometric measurements, which enable a coarse taxonomic classification, an important step for estimating albedo. To ensure that the inputs are both easily accessible and sufficiently informative, we construct the dataset based on absolute magnitude, type information, and orbital parameters. By removing the reliance on spectroscopy, the available sample size increases substantially, facilitating the training of a more robust deep learning model. Furthermore, by optimizing the model to adapt to the available inputs, we aim to achieve inversion accuracy comparable to that of methods requiring Gaia spectra.

The labels for asteroid albedo and diameter are primarily derived from the NEOWISE Diameters and Albedos v2.0

dataset (A. Mainzer et al. 2019), based on infrared observations from the Wide-field Infrared Survey Explorer (WISE)/NEOWISE mission (J. R. Masiero et al. 2011). In addition, several radar measurements of NEAs replaced the corresponding NEOWISE values to improve the reliability. The WISE space telescope is equipped with four wavelength bands at 3.4, 4.6, 12, and 22 μm and conducted observations in 2010 for numerous asteroids. After the cryogenic coolant was exhausted, it was reactivated in 2013 as NEOWISE and has continued to the present, collecting mid-infrared data for small bodies using only the 3.4 and 4.6 μm bands. Both mission phases employed the NEA thermal model to derive size and albedo parameters of asteroids. Although the NEOWISE-phase results exhibit higher uncertainties, their reliability remains superior to estimates based solely on optical observations (J. R. Masiero et al. 2011; N. Myhrvold 2018).

The uncertainty statistics of the NEOWISE dataset are shown in Figure 1. For the 74,185 samples used, the mean relative uncertainties are 34.5% for albedo and 12.8% for diameter, serving as the benchmark for evaluating model performance and the fidelity of the resulting asteroid property catalog.

Asteroid type is an important feature for estimating albedo and diameter. The Solar System Open Database Network (SsODNet; J. Berthier et al. 2023) provides a compilation of existing asteroid parameters from multiple sources. The type results from SsODNet are merged into six primary types (A, C, D, S, V, X) to avoid confusion among different classification schemes, and samples with unassignable types are removed. To further expand our sample size, we integrate results from the Asteroid Physical Parameters–Gaia DR3–AI Prediction (AGAI) catalog (J. Ge et al. 2025), which are classified based on Gaia DR3 spectra, retaining only those with a classification confidence greater than 0.5 to ensure reliability. Meanwhile, to guarantee the physical plausibility of the target values, samples with an albedo greater than 0.7 are excluded as outliers.

In addition to real samples, data augmentation was performed while preserving both diversity and realism. First, we generated simulated data for 200 NEAs to enrich the representation of small near-Earth objects in the dataset. This was done by randomly selecting NEAs fainter than magnitude 18, assigning types according to the statistical proportions of real samples, and calculating diameters using the Bowell relation based on the mean albedo for each type. Furthermore, Gaussian noise augmentation was applied to each parameter of the real samples based on its statistical error level. Subsets with fewer samples in specific size bins were subjected to more augmentation to alleviate the imbalance. The number of augmented samples is deliberately limited to avoid feature homogeneity, since the uncertainties in orbital parameters are relatively small.

After these procedures, the resulting dataset contains 143,552 samples, of which 69,269 are real samples, providing sufficient data for subsequent model training and validation. The distributions of the key features in the dataset are presented in Figure 2. The absolute magnitudes of these asteroids approximately follow a normal distribution, with a mean value around 15. Regarding taxonomic types, C- and S-type asteroids dominate the dataset, whereas A- and V-types are relatively scarce, in agreement with current understanding of the solar system asteroid population. Diameters are primarily concentrated in the 1–10 km range, with 628

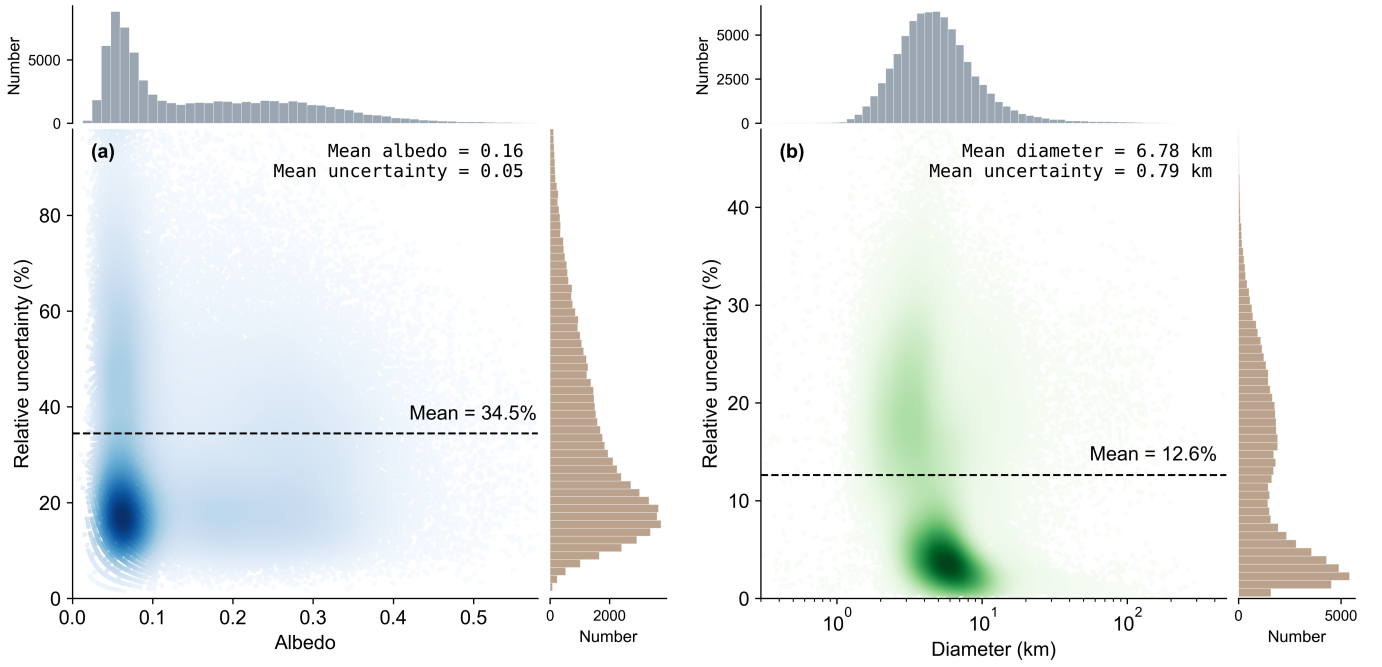


Figure 1. Uncertainty statistics of the NEOWISE dataset. The relative uncertainty is defined as the ratio of the reported uncertainty to the estimated value itself.

asteroids smaller than 1 km, 8760 exceeding 10 km, and 954 NEAs. Albedo values for all samples lie within the range of 0.02–0.7.

For dataset partitioning, asteroids visited by spacecraft are reserved as an independent test set to ensure an objective evaluation of model performance. The remaining samples are split into training and validation sets in a 9:1 ratio. For the final model used to construct the catalog, these in situ samples are reintegrated into the training set to maximize the generalization capability.

2.2. Input Information

The model’s ability to accurately map the inputs to the target parameters fundamentally depends on whether the provided features are sufficiently informative. We detail the multisource input information included in the dataset, as shown in Table 1. The orbital parameters comprise seven elements, such as semimajor axis and eccentricity, obtained from the Minor Planet Center (MPC), while the physical parameters consist of four key quantities, including absolute magnitude and type related information, sourced from SsODNet and the AGAI catalog.

A notable feature of this dataset is the introduction of an additional dimension termed type quality, which is designed to represent the reliability of the taxonomic type. It is classified into three levels:

Level 1 indicates low reliability, with asteroid type determinations based on multiband photometry or simply inferred from orbital information.

Level 2 represents moderate reliability, with asteroid types derived from visible or near-infrared (NIR) spectra or from low-resolution spectral data such as those in Gaia DR3.

Level 3 denotes high reliability, referring to asteroid types determined from visible to near-infrared (VisNIR) spectroscopy (e.g., 0.45–2.45 μm) or validated through radar or other reliable measurements.

2.3. Asteroid Albedo-diameter Inversion Model

J. Ge et al. (2025) proposed the AadRF model, which combines asteroid spectra and selected parameters using deep learning and random forest, achieving more robust performance in asteroid albedo and diameter estimation compared to other methods. Building on this, we developed a spectral-free asteroid albedo-diameter joint inversion network, AadNet2, as shown in Figure 3. It takes an 11-dimensional feature vector of easily accessible yet essential parameters as input and simultaneously outputs the asteroid’s geometric albedo and effective diameter.

As for the model architecture, the raw data are first processed through a three-layer fully connected network for feature extraction. These layers follow an hourglass-shaped structure, expanding and then compressing the feature space to obtain a 64-dimensional high-level semantic representation. Each layer employs the LeakyReLU (J. Xu et al. 2020) activation function and incorporates a dropout rate of 0.2 for regularization to prevent overfitting. Building on the extracted features, a self-attention (A. Vaswani et al. 2017) module is introduced. It consists of 64 attention heads, each independently learning correlation patterns among feature dimensions. The attention mechanism allows the model to adaptively prioritize different features, thereby identifying their most discriminative combinations.

AadNet2 inherits the prediction strategy of the AadRF model, in which albedo is first predicted to provide guidance for the diameter prediction. The albedo-prediction branch progressively maps the 64-dimensional feature representation to an albedo estimate, with the absolute value applied to enforce physical plausibility. Next, the Bowell formula (Equation (1)) embedded in the network is employed to compute a diameter reference value, which is subsequently concatenated with the previously extracted feature to form an enhanced 65-dimensional feature vector. Finally, the diameter-prediction branch takes the fused feature as input to produce the final diameter estimate. This design leverages the intrinsic

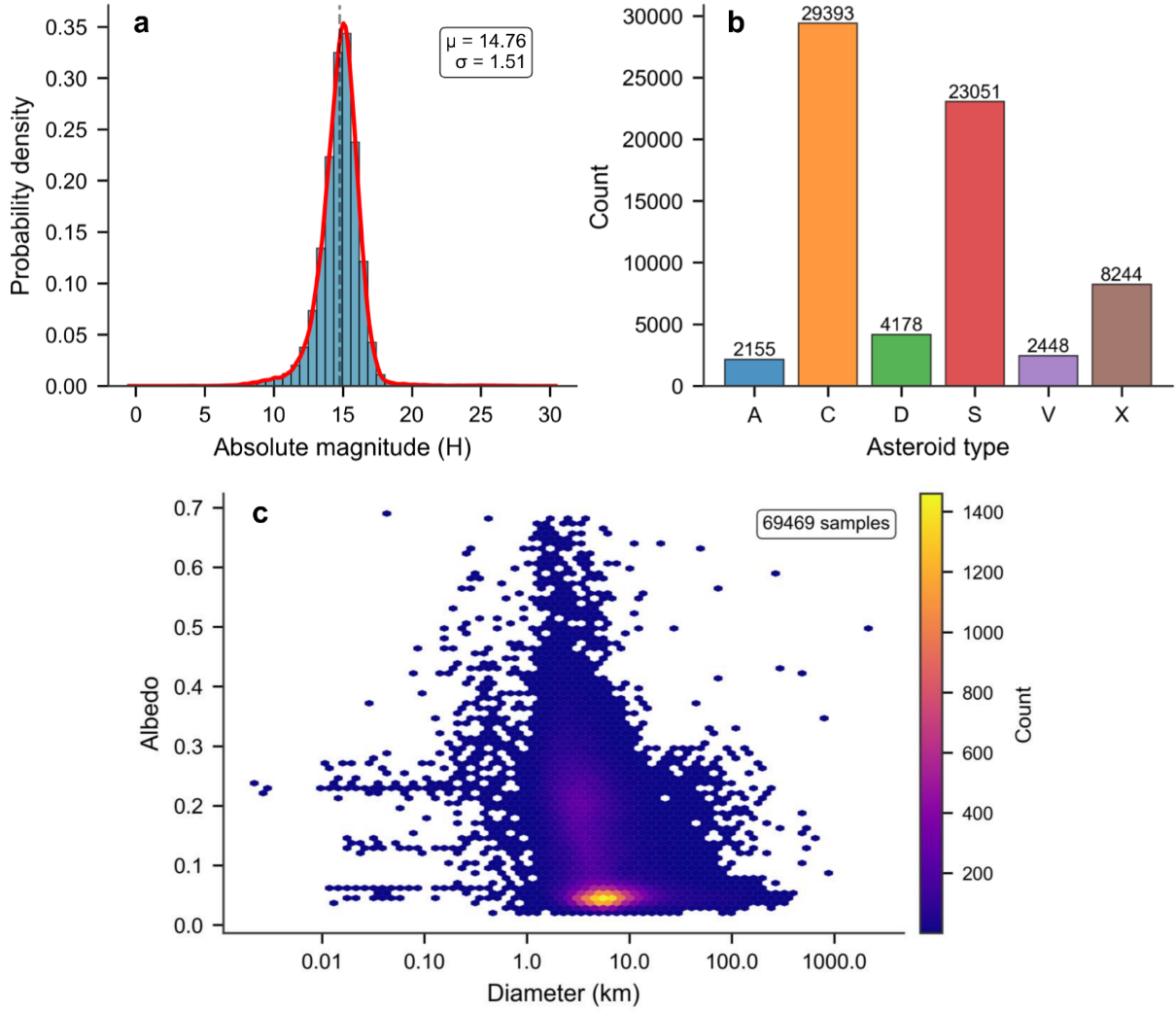


Figure 2. Distribution of asteroid properties in the dataset.

Table 1
Input Information for Asteroid Albedo-diameter Inversion

Data	Specific	Source
Orbital parameters	a	MPC
	e	
	i	
	q	
	Q	
	Orbital period Semilatus rectum	
Physical parameters	H	MPC
	Type	SsODNet, AGAI
	Type quality	Classification method
	Initial albedo	Statistical

Note. a —semimajor axis; e —eccentricity; i —inclination; q —perihelion distance; Q —aphelion distance; H —absolute magnitude. Initial albedo refers to the average albedo value for each asteroid type.

coupling between an asteroid’s albedo and its size, which contributes to improving the robustness of the model. In addition, due to the large volume of training data in this study, the advantages of deep learning become more evident.

To enhance the practical convenience of the approach, random forests are no longer integrated.

$$D = \frac{1329}{\sqrt{p_V}} \times 10^{-H/5} \quad (1)$$

where D is the effective diameter, p_V is the albedo, and H is the absolute magnitude.

To ensure that the model remains applicable and operationally reliable, necessary constraints are imposed during the inference stage. When the network’s predictions fall outside valid ranges, the system automatically reverts to a conventional method to ensure the validity of the results. Specifically, the valid output range for albedo is set to 0.02–0.7. Predictions outside this range are recalculated using the Bowell formula based on the model-predicted diameter. Regarding the diameter, its valid range is set to 0.01–1000 km. If the predicted value falls outside this interval, the diameter is derived using the Bowell formula based on the predicted albedo. This strategy is adopted because extremely small asteroids, which are only a few meters in size, are prone to substantial errors due to the lack of such samples. If both albedo and diameter estimates fall outside their valid bounds, which is very rare, the system defaults to using the type-average albedo and calculates the diameter in the same way.

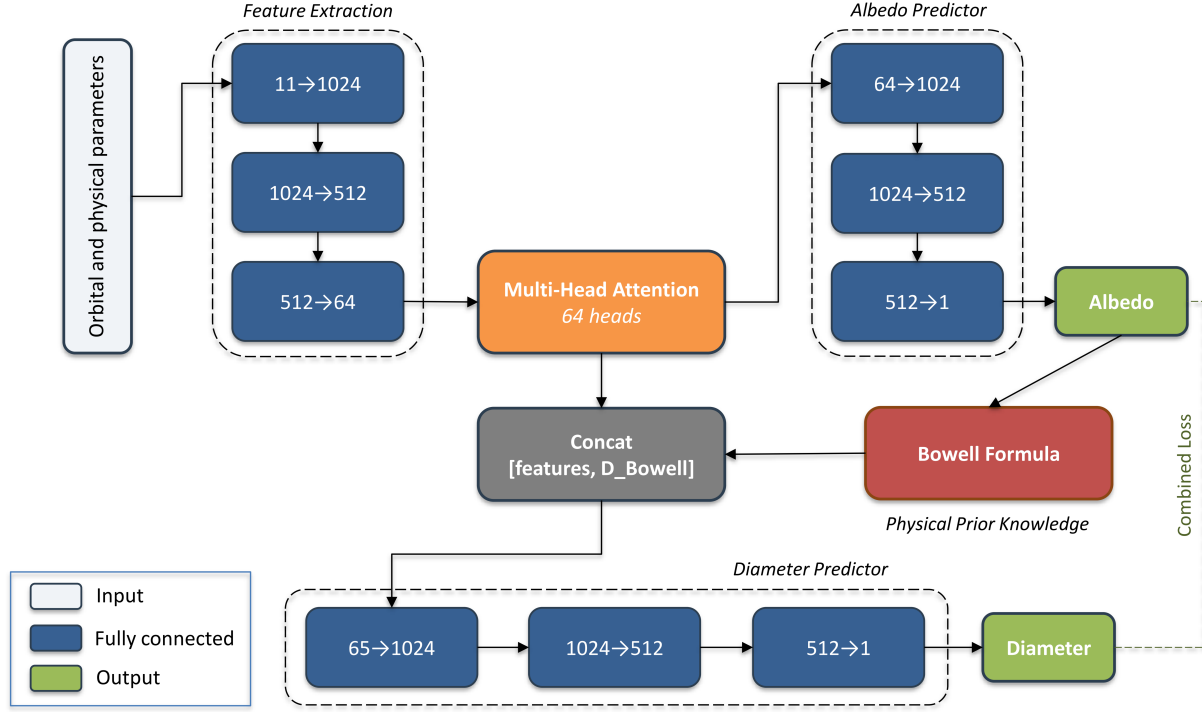


Figure 3. Model architecture of AadNet2.

During model training, mean squared error is employed as the loss function for both albedo and diameter. The total loss is defined as a weighted combination of the two components:

$$\text{Loss} = 0.6 \cdot L_A + 0.4 \cdot L_D \quad (2)$$

where L_A and L_D denote the losses for albedo and diameter, respectively. The weights are optimized through multiple rounds of experimentation, which highlights the fundamental importance of albedo prediction. During inference, all input values should be standardized using Z-score normalization, applying the same means and standard deviations estimated from real samples during the training phase. The hyperparameter settings of the model are summarized in Table 2. Under this configuration, the training process exhibit stable convergence. The model is implemented using the PyTorch 2.3.0 framework (J. Ansel et al. 2024) and trained on an NVIDIA GeForce RTX 4090 GPU.

3. Method Evaluation

We compared the accuracy of the relevant methods with our model on the validation set. Among them, the *Bowell* relation refers to the traditional method. AadRF is the current state-of-the-art method using spectral data and provides the conceptual foundation for the AadNet2 model. It should be noted that AadRF and AadNet2 employ different input features and training datasets, so their metrics are not strictly comparable but remain informative. In addition, to assess the necessity of type quality as an input, an ablation experiment is conducted by removing this feature, yielding the AadNet2-NoTQ variant. Its training dataset and hyperparameter configuration are kept identical to those of AadNet2.

The performance of the various methods is summarized in Table 3. It can be seen that, relative to the *Bowell* relation, AadNet2 achieves a 47.8% reduction in albedo root mean

Table 2
Hyperparameters of AadNet2 Model

Hyperparameter	Value
Learning rate	1×10^{-5}
Batch size	64
Optimizer	AdamW
Weight decay	1×10^{-4}
Epochs	2000
Dropout	0.2

square error (RMSE) and a 62.9% reduction in diameter RMSE, reaching 0.047 and 3.293 km, respectively. This demonstrates the advantage of multiparameter fusion using neural networks. Compared with the AadNet2-NoTQ model, the inclusion of type quality leads to an enhancement in the accuracy of target properties. This indicates that such information assists the model in evaluating the reliability of asteroid classifications, thereby providing better constraints for the inversion results. Relative to AadRF, AadNet2 achieves a similar level of accuracy in albedo prediction. Regarding diameter, AadNet2's higher RMSE is primarily driven by the absence of rotational period constraints, whereas its lower mean absolute error stems from a training distribution skewed toward smaller asteroids. Overall, AadNet2 yields comparable accuracy to AadRF without requiring spectral data and extensive parameters like rotational periods. This is primarily because removing constraints on inputs that are not readily obtainable allows for the construction of a much larger dataset, facilitating the training of a model with strong generalization performance.

To further assess the performance of AadNet2 against reference values from the most reliable sources, we tested the model on 16 asteroids that have been visited by spacecraft or observed via radar, with the results summarized in Table 4. It

Table 3
Comparison of Validation Accuracy across Different Methods

Method	Albedo		Diameter (km)		Required Inputs
	RMSE	MAE	RMSE	MAE	
Bowell relation	0.090	0.071	8.882	2.025	H , type
AadRF	0.049	0.032	3.232	1.482	Spectra, orbital and physical parameters
AadNet2-NoTQ	0.050	0.032	3.423	0.967	Orbital and physical parameters
AadNet2	0.047	0.030	3.293	0.929	

Table 4
Test Results of the AadNet2 Model

MPC Number	Albedo			Diameter (km)		
	Predicted	Truth	Error (%)	Predicted	Truth	Error (%)
4	0.27	0.42	35.7	563.93	525.4	7.3
16	0.08	0.12	33.3	322.70	222	45.4
21	0.23	0.22	45.5	89.96	95.76	6.1
243	0.21	0.21	0.0	28.31	31.4	9.8
253	0.06	0.04	50.0	54.63	52.8	3.5
433	0.20	0.25	20.0	21.58	16.84	28.1
951	0.28	0.25	12.0	13.53	12.2	10.9
2867	0.21	0.30	30.0	7.63	5.16	47.9
4179	0.23	0.16	43.8	2.42	2.73	11.4
25143	0.23	0.27	14.8	0.39	0.33	18.2
33342	0.19	0.30	36.7	0.86	0.60	43.3
38071	0.06	0.06	0.0	0.66	0.64	3.1
68347	0.34	0.30	13.3	0.30	0.25	20.0
101955	0.07	0.04	75.0	0.34	0.48	29.2
162173	0.06	0.05	20.0	0.66	0.90	26.7
418849	0.24	0.29	17.2	0.27	0.20	35.0
MAPE	26.4%			21.6%		

is observed that the errors of albedo predictions are mainly controlled within 50%, with a maximum of 75.0%. This peak error occurs for the carbonaceous asteroid 101955 (Bennu), primarily because its small ground-truth albedo leads to a high percentage error despite its absolute error being small. In contrast, Vesta (4) exhibits the largest absolute error, which is attributed to its high albedo as a typical V-type asteroid, whereas samples with albedo above 0.4 are relatively rare in the dataset. A similar situation occurs for asteroid 2867 (Šteins), where the model underestimates its albedo, leading to an overestimation of the effective diameter with an error of 47.9%, the highest among the test samples. Asteroids with more accurate albedo predictions generally show better diameter estimates, indicating that the model effectively leverages the intrinsic coupling between them.

Overall, the model achieves MAPEs of 26.4% and 21.6% for albedo and diameter prediction, respectively, indicating that diameter estimation is slightly more accurate than that of albedo. In comparison, the spectrum-based AadRF model yields MAPEs of 28.9% and 17.5% for the two parameters. The variations in accuracy between the two models are expected because of the sample differences. In addition, AadNet2 performs better for smaller asteroids and exhibits less variability in prediction errors, benefiting from the improved generalization stability afforded by the larger dataset. This demonstrates that a model relying only on orbital parameters, absolute magnitude, and type information can achieve

performance comparable to one requiring spectral data, highlighting its broader practical applicability.

4. Asteroid Physical Properties Catalog

After validating the method, we applied the AadNet2 model to predict the albedo and diameter of all asteroids with known types. Based on the prediction results, we constructed the Asteroid Type, Albedo, and Diameter (ATAD) catalog, followed by accuracy evaluations and analyses. Except for the type and its quality flag, all other input parameters are sourced from the MPC.

4.1. Overview

The generated ATAD catalog contains 188,737 asteroids and provides three fundamental physical properties: type, geometric albedo, and effective diameter. The fields of this catalog are listed in Table 5. Specifically, the types are taken from the SsODNet and AGAI classifications, while albedo and diameter are predicted by the AadNet2 model.

The types from SsODNet are primarily derived from multiband photometric data, whereas the AGAI catalog delivers classifications based on Gaia spectra via the AadRF model, accompanied by classification confidence. The `type_method` field includes seven categories, indicating whether the classification is derived from multiband photometry, spectra, or AI predictions using Gaia DR3 spectra. The photometric and spectral (non-Gaia DR3) classifications are further divided into three wavelength regimes: visible light (VIS), NIR, and VisNIR. Among these sources, photometric results exhibit the least reliable, spectra are the most definitive, and AI predictions fall in between due to the limited spectral resolution and the possibility of misclassification. Moreover, results based on VisNIR spectra are considered more reliable than those derived from VIS or NIR alone. The `type_method` attribute not only serves as a provenance for the types within the catalog but also defines the type quality value used as input to AsterNet2.

The uncertainties of albedo and diameter are calculated by considering both the deviation of predictions from the NEOWISE dataset and the inherent uncertainties of NEOWISE itself, using the following formulas:

$$\sigma_{p_V} = p_V \cdot \sqrt{\text{MAPE}_{p_V}^2 + \delta_{p_V, \text{NEOWISE}}^2} \quad (3)$$

$$\sigma_D = D \cdot \sqrt{\text{MAPE}_D^2 + \delta_{D, \text{NEOWISE}}^2} \quad (4)$$

where σ_{p_V} and σ_D are the uncertainties of albedo and diameter, respectively; p_V and D are the predicted albedo and diameter; MAPE_{p_V} is the mean albedo-prediction error relative to NEOWISE for the corresponding type, and MAPE_D is the mean diameter-prediction error relative to NEOWISE for the

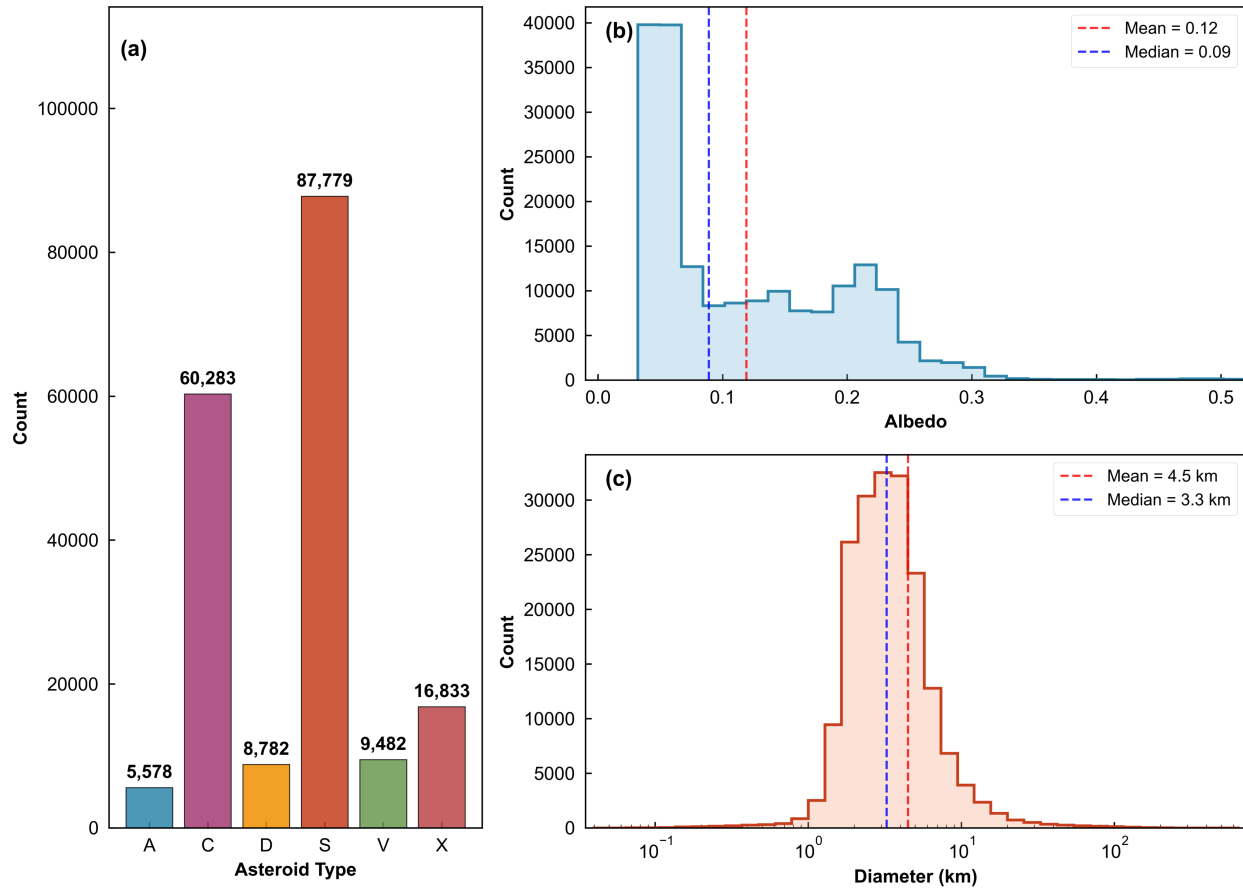


Figure 4. Distributions of asteroid properties in the ATAD catalog.

Table 5
Fields of the ATAD Catalog

MPC Number	Type	Type_method	Type_confidence	Albedo	σ_{p_V}	Diameter (km)	σ_D (km)
2	C	Reflectance spectroscopy (VisNIR)	...	0.096	0.044	545.462	84.202
215	S	AI predicted (Gaia DR3)	0.68	0.208	0.100	35.813	6.642
277	S	Reflectance spectroscopy (VIS)	...	0.199	0.096	31.533	5.848
296	S	Multifilter photometry (VIS)	...	0.247	0.119	8.209	1.469
370	X	Multifilter photometry (NIR)	...	0.065	0.034	41.486	7.694
2012	C	Multifilter photometry (NIR)	...	0.054	0.025	10.956	2.032
20461	D	Multifilter photometry (VisNIR)	...	0.039	0.020	9.776	1.749
90966	A	AI predicted (Gaia DR3)	0.88	0.206	0.092	1.920	0.383
313962	C	AI predicted (Gaia DR3)	0.44	0.040	0.018	5.233	0.936
2014 WE120	V	Reflectance spectroscopy (VisNIR)	...	0.399	0.177	0.116	0.052

Note. Type_confidence refers to the classification confidence provided in the AGAI catalog. σ_{p_V} and σ_D denote the uncertainties of albedo and diameter, respectively.

(This table is available in its entirety in machine-readable form in the [online article](#).)

corresponding size bin, as detailed in Figure 5. $\delta_{p_V, \text{NEOWISE}}$ and $\delta_{D, \text{NEOWISE}}$ are the mean uncertainties from the NEOWISE dataset as shown in Figure 1, with values of 34.5% and 12.6%, respectively.

Figure 4 shows the distribution of asteroid properties in the ATAD catalog. S-types are the most abundant, with nearly 88,000 objects representing 46.5% of the total sample, followed by C-type asteroids. The population of X-types is also considerable. Following M. Delbo et al. (2019), the

X-complex can be further divided based on albedo (p_V) into P-type ($p_V < 0.1$), M-type ($0.1 \leq p_V \leq 0.3$), and E-type ($p_V > 0.3$), yielding a total of eight classes. The abundances of V-type and D-type asteroids are comparable, while A-type asteroids are the least common, indicating that bodies with pronounced olivine signatures are relatively rare.

The statistical results also show that 52.8% of the asteroids in this catalog have albedos below 0.1, reflecting the high abundance of primitive asteroids such as C-type and D-type. A

Table 6

Comparison of Sample Counts between the ATAD and AGAI Catalogs

Criteria	AGAI	ATAD
Diameter < 1 km	27	2486
Diameter < 2 km	2127	34,011
Diameter < 5 km	30,818	147,745
NEAs	97	2447

secondary peak appears near 0.24, corresponding to S-type and A-type. Approximately 2600 asteroids in the ATAD catalog exhibit albedos exceeding 0.3, and the mean albedo for the entire catalog is 0.12. Regarding size distribution, the vast majority of asteroids fall within the 1–10 km range, with an average size of 4.5 km. Compared with the AGAI catalog, the ATAD catalog achieves higher completeness, with a substantial increase in its coverage of small asteroids and near-Earth objects. A detailed comparison is provided in Table 6. It should be emphasized that the parameter distribution of samples in this catalog differs slightly from that of SsODNet, which is mainly reflected in reduced consistency for the extreme parameter ranges of albedo and diameter. Therefore, the above statistical results do not constitute a rigorous mapping of the true population distribution of asteroids.

4.2. Catalog Accuracy

The overall accuracy of asteroid albedo and diameter in the ATAD catalog is shown in Figure 5. The evaluation relies on 74,191 samples matched with the NEOWISE dataset.

For albedo predictions (Figure 5(a)), different types exhibit notable variations in accuracy. V-type and A-type asteroids show the best performance, with MAPE values of 27.8% and 28.5%, respectively. This can be attributed to the distinct and consistent characteristics of these types, which enable the model to effectively capture the relationship between taxonomy and albedo. In contrast, X-type exhibits the highest prediction error, reaching 40.0%, highlighting the greater heterogeneity and complexity within this group.

Diameter-prediction errors (Figure 5(b)) follow a size-dependent trend. The model achieves superior accuracy for large asteroids, with MAPE values below 10% for objects exceeding 50 km in diameter. For medium-sized asteroids in the 1–50 km range, the error remains within 20%. However, accuracy decreases noticeably for bodies in the 0–1 km range, where the MAPE reaches 42.6%. This reflects not only the intrinsic difficulty of inferring the parameters of small objects, but also the relatively limited number of samples available within this size range.

The statistics in panels (c) and (d) reveal systematic patterns in model prediction errors, which are related to the uneven distribution of training samples across different intervals. The overall albedo MAPE of the catalog is 33.1%, representing a reduction of 40.9 percentage points compared with the traditional type-averaged method. The diameter error distribution also exhibits dispersion at the small-size end, with an overall MAPE of 14.4%, which is 10.6 percentage points lower than the result calculated by the Bowell relation. These statistical results provide guidance for future optimization.

In addition, compared with the AGAI catalog, in terms of accuracy relative to NEOWISE results, the ATAD catalog shows a reduction of 3.7 percentage points in albedo error and a reduction of 1.7 percentage points in diameter error. Given

that the datasets underlying the two catalogs differ in sample size, quality, and distribution, these variations in accuracy are reasonable. Combining validation accuracy with test results, there is sufficient evidence to conclude that the AadNet2 model performs comparably to the Gaia spectrum-based AadRF model, while the ATAD catalog is substantially larger in scale. Therefore, the ATAD catalog constructed by the proposed deep learning approach demonstrates substantial practical value.

To further validate the rationality of the sample distribution in the ATAD catalog, we compared it with the matched samples from the SsODNet database. As illustrated in Figure 6, the vast majority of data points cluster closely along the 1:1 reference line, demonstrating that the model delivers stable diameter-prediction performance for asteroids of typical sizes. For objects smaller than 1 km in diameter, the values from ATAD tend to lie above the reference line. This overestimation arises from the model’s lack of sufficient training data to learn the intrinsic features of small-sized asteroids, which drives the model to inflate predictions for the purpose of loss reduction. Furthermore, as illustrated in Figure 7, the two datasets exhibit high consistency in the morphology and coverage of their core regions, with some discrepancies in the marginal ranges. Specifically, the ATAD catalog contains few samples with albedos below 0.04, and the number of samples at the high-albedo end above 0.35 is also lower than that in SsODNet. This still arises from the insufficient representation of asteroids in these albedo ranges in the training data, leading to inadequate feature learning. Additionally, the loss reduction process causes the model to regress toward the mean of the training set. In addition, the discontinuous feature at the low-albedo end is associated with the similar sample distribution in the NEOWISE dataset.

4.3. Discussion

AadNet2 places emphasis on practicality and is optimized to achieve high performance without relying on spectra, thereby enabling broader applicability. The enhanced generalization ability of AadNet2 is primarily attributable to the expansion of data scale and the incorporation of type quality information.

As a pioneering application of machine learning to asteroid characterization, the AGAI catalog serves as the foundation for building the ATAD catalog by providing extensive asteroid taxonomic data. The developed ATAD catalog maintains comparable accuracy in physical properties while offering significantly greater sample coverage. This catalog could facilitate scientific research and engineering missions. For example, after crossmatching the ATAD catalog with orbital parameters from the MPC database, interested candidates can be screened as needed according to physical and orbital parameters. In this procedure, the catalog serves to greatly narrow down the potential target range. Subsequently, high-resolution spectroscopic observations can be arranged, and cross validation can be performed with other independent research results to confirm the final targets. In addition, the significance of constructing the synthetic catalog is to deliver comparative results, precision benchmarks, and methodological references to support the further development of deep learning-based asteroid property inversion.

It should be noted that the three fundamental properties provided by the ATAD catalog overlap with established records in existing databases. Among these, results obtained

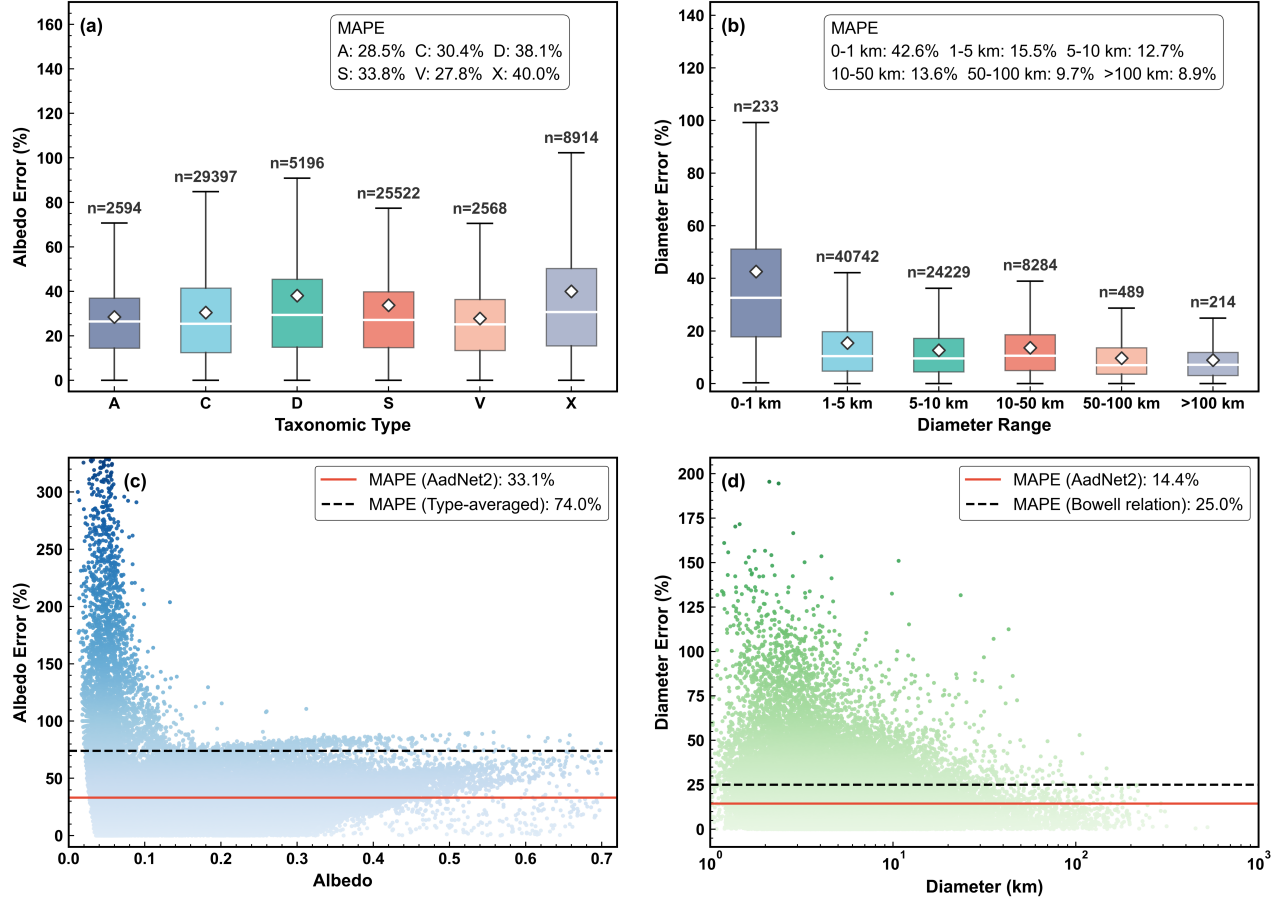


Figure 5. Parameter accuracy of the ATAD catalog in comparison with the NEOWISE dataset.

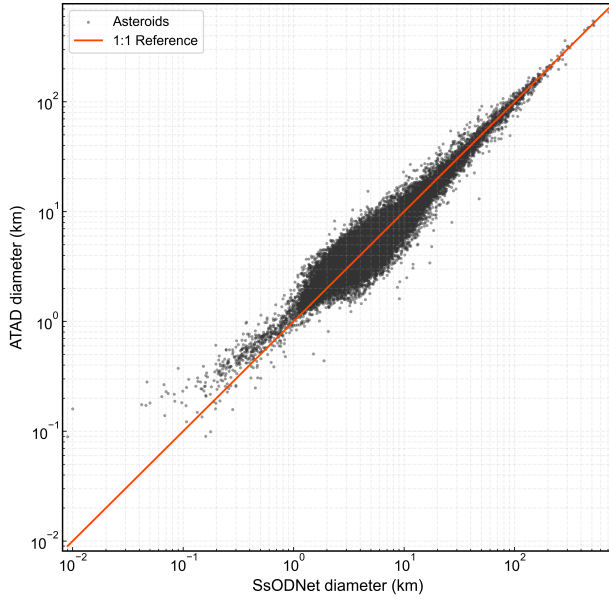


Figure 6. Comparison of asteroid sizes between ATAD and SsODNet.

through reliable detection methods such as infrared and radar observations are generally prioritized. Based on the albedo and diameter distribution within the ATAD catalog, we define the valid parameter ranges as 0.04–0.7 for albedo and 1–1000 km for diameter, applicable to both the ATAD catalog and the

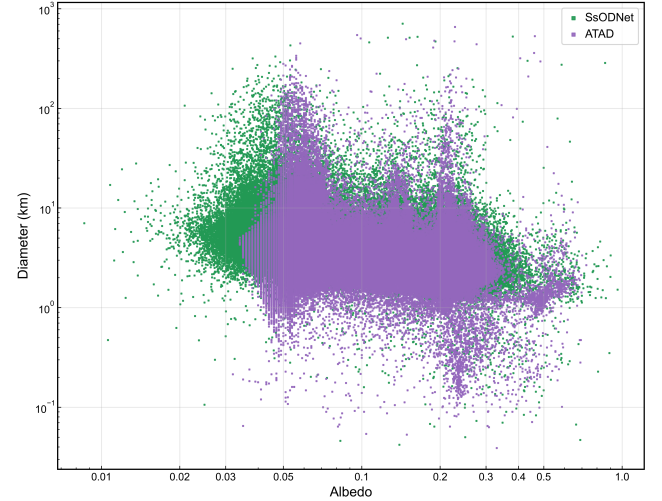


Figure 7. Comparison of parameter distributions between ATAD and SsODNet.

AadNet2 model. Predicted values outside these ranges have limited reliability and are not encouraged for use. For asteroids with predicted diameters below 1 km, we recommend recalculating the effective diameter using Bowell's relation with estimated albedo of AadNet2. In addition, the current catalog and model are not applicable to dwarf planets, multiple systems such as binary asteroids, and active asteroids. For objects with highly irregular shapes, effective diameter estimates also tend to exhibit greater uncertainties.

5. Conclusion

This paper proposes a deep learning framework that integrates widely accessible and diagnostic features, including absolute magnitude, orbital elements, and taxonomic information from optical-band observations to construct a comprehensive catalog of asteroid fundamental physical properties. The developed AadNet2 model obviates the need for inputs with limited coverage such as spectral data, thereby substantially increasing the dataset scale to ensure robust generalization. Test results on objects that have been explored by spacecraft or radar observations show that the model achieves MAPE values of 26.4% and 21.6% for albedo and diameter, respectively.

Based on the taxonomic classifications provided by the SsODNet database and the AGAI catalog, the AadNet2 model was applied to predict geometric albedo and effective diameter for all collected asteroids with known types. The resulting ATAD catalog contains type, albedo, and size for 188,737 asteroids, achieving significantly improved completeness over the previous AGAI catalog while maintaining comparable accuracy. We have delineated the practical applications of this work and released the catalog to support the broader research community.

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Data Availability

The generated ATAD catalog is available on Zenodo at doi:[10.5281/zenodo.20032293](https://doi.org/10.5281/zenodo.20032293), and also as a machine-

readable file associated with Table 5. Additional training data is made available on Zenodo.

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